# DSAII- HW3

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## Problem 1

Prove there is always a starting post that ensures the cat will win. Prove by Induction on $n$ for $n \geq 1$.
**BC**: There are 2 posts, $n = 1$. In this case there is 1 Red and 1 Black post.
In the case the cat starts on the Red post, then trivially, it has visited at least 0 red posts ($1\ Red \geq 0\ Black$). Then when the cat goes to the black post, the cat wins since $1\ Red \geq 1\ Black$. Thus when the cat starts on red the base case is always True.
**IH**: The can can always win in the case where the input has at most $2(n-1)$ posts. (From Problem Statement)
**IS**:
If there are $2n$ posts then there must be a pair of Red and Black posts that is adjacent. If we remove these 3 posts there are $2(n-1)$ total posts with there being $(n-1)$ Red and $(n-1)$ Black. By the IH, this pattern must have Winning starting position.
If the smaller pattern has a winning start, then the 2 posts we took away earlier doesn't effect the Smaller pattern since the base case is True, thus there is always a winning position.

## Problem 2

Prove by Induction that any fully parenthesized product of Matrices has exactly $n-1$ pairs of parentheses.
**BC:** In the case that we have a single product of matrices, $n = 2$. Trivially, this satisfies the proof that there are exactly $n-1$ pairs of parentheses, thus True.
**IH:** For any fully parenthesized product of $k$ matrices, where $k < n$ there are exactly $k-1$ pairs of parentheses.
**IS:** In this case we must have $n > 2$ matrices. For the product of $n$ matrices we have the format $(AB)$ where A is the product of $i$ matrices for some $1 \leq i < n$ and B is the product of $j$ matrices where $j = n - i$. This the total amount of matrices being multiplies is $n = i + j$.
By the IH, A must have $i-1$ parentheses and B must have $j-1$ parentheses. When counting the total amount of parentheses for the entire product, $P$ we get $P = (i-1) + (j-1) + 1$.

The 1 is for the pair that wraps ($AB$), thus $P = i + j - 1 = n - 1$, so this is true.

## Problem 3

**Part A**

For the algorithm, we need to 1st find a Knight. If a knight always tells the truth, we can identify all the others using the knight. Once we have the knight, we only need $n - 1$ questions to identify the others.

```
findKnight(people)
    if len(people) = 1
        return people[0] //Must be the knight
    winners = [] //Now pair up N people
    for each pair (A,B):
        Ask A: "what is B's type"
        if A answers "B is a knight"
            add B to winners
        else
            Discard Both A and B

    return findKnight(winners) //recursivly search through possible knight
```

In this structure there are 4 cases when we ask about the other in the pair.
1. If A is a knight and answers B is a knight, B is a knight so we keep B.
2. If A is a knight and answers B is a knave, we discard both.
3. If A is a knave and answers B is a knight, B is unknown but we keep them and we discard A.
4. A is a knave and answers B is a knave, We lose both and B's identity doesn't matter.

We only know we loose a Knight in Case 2, where we also loose a Knave, since we started with majority knights, this is maintained.
Once we have the knights for certain, we can identify the rest with the knight.

**Part B**

Claim: If at most $\frac{n}{2}$ people are knights, it is impossible to determine everybody's type. Thus, the number of knight must be $n \leq \frac{n}{2}$.

Proof by Counter Example
Suppose we have 4 people, 2 knights and 2 knaves. We will call the Knight $A$ and $B$ and the knaves $C$ and $D$.
Knights $= A, B$
Knaves $= C, D$
If the knaves always tell the truth, we cannot identify anybody since because at least 2 of the answers are uncertain, we cannot identify anybody.

On the contrary if there is only 1 knave and 3 knights (majority knight), The knave's response doesn't matter since there are 3 consistent answers that identify the knave. So there must be majority knights to identify everybody.

## Problem 4

**Part A**
For the recurrence relation, The initial Base Case is that the red ball starts in the the 1st bin (left), so $p(1) = 1$. To find the recurrence $p(n+1)$, we need to find the probability that the red ball is in bin q at the start of the next round.
Case 1: The red ball is in bin 1 at the start of round $n$. For the red to stay in the bin there is a $\frac{3}{4}$ chance.
Case 2: The red ball is in bin 2 (right) at the start of round n. For the red to move back to bin 1 after the swap, we must pick it our from the other bin. This has a $\frac{1}{4}$ chance of happening. This is the same as $1 - p(n)$
So the recurrence relation is $p(n+1) = p(n) \cdot \frac{3}{4} + (1 - p(n)) \cdot \frac{1}{4}$
Simplifying this we get the recurrence relation $p(n+1) = \frac{1}{2} \cdot p(n) + \frac{1}{4}$

**Part B**
To make a good guess we are going to step through n till we see a pattern.
$p(1) = 1$
$p(2) = \frac{1}{2}p(1) + \frac{1}{4} = 3/4$
$p(3) = \frac{1}{2}p(2) + \frac{1}{4} = 5/8$
$p(4) = \frac{1}{2}p(3) + \frac{1}{4} = 9/16$

The patterns that emerges is $p(n) = 2^{n-1} + 1/2^n$

Proof by induction on n
**BC:** $n = 1$. $p(1) = 1$ is given, in our equation, $p(1) = 2^0 + 1/2 = 1$, hence this is correct.
**IH:** For $n \geq 1$, $p(n) = 2^{n-1} + 1/2^n$
**IS:** For $n + 1$
$p(n+1) = 1/2 \cdot p(n+1) + 1/4$
$p(n+1) = 1/2(2^{n-1} + 1/2^n) + 1/4$
$p(n+1) = 2^{n-1} + 1/2 \cdot 2^n + 1/4$
$p(n+1) = 2^{n-1} + 1/2^{n+1} + 1/4$
We can convert $1/4$ to have a common denominator so $1/4 = 2^{n-1}/2^{n+1}$
$p(n+1) = 2^{n-1} + 1/2^{n+1} + 2^{n-1}/2^{n+1}$
$p(n+1) = 2^{n-1} + 1 + 2^{n-1}/2^{n+1}$
$p(n+1) = 2 \cdot 2^{n-1} + 1/2^{n+1}$
$p(n+1) = 2^{n+1} + 1/2^n$
Thus this is correct.
**Part C**
Let $X_i$ be an indicator variable where $X_i = 1$ if the red ball being in bin 1 at the start of round $i$, 0 otherwise.
The total number of rounds the ball is expected to be in bin 1 is $X_1 + X_2 + ...X_m$.

By LOE we can get the expectation
$E[X_1 + X_2 + ... + X_m] = E[X_1] + E[X_2] + ... + E[X_m]$
$\sum_{i=1}^{m} X_i$
$X_i$ is the same as our result from part b, so $X_i = 2^{n-1} + 1/2^n$
$\sum_{i=1}^{m} 2^{i-1} + 1/2^i$
$\sum_{i=1}^{m} 2^{i-1}/2^i + \sum_{i=1}^{m} 1/2^i$
The 1st sum
$\sum_{i=1}^{m} 2^{i-1}/2^i = m/2$ The 2nd sum
$\sum_{i=1}^{m} 1/2^i = 1 - 1/2^m$
Putting the sums back together we get
$E[\text{Total rounds in Bin 1}] = m/2 + (1 - 1/2^m)$